Computer Science & IT

COMPUTER NETWORKS (CN)

Error Control Methods

One Shot



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Recap of Previous Lecture



- Topic
- Topic
- Topic

FC, AC

Topics to be Covered



- Topic
- Topic
- Topic

E C



Inspiring Stories : Veer Narayan Singh

Veer Narayan Singh Background: Born in 1795 in Chhattisgarh, a respected farmer and community leader.

Struggle: Witnessed British officers hoarding grain while villagers starved during famine.

Education: Learned leadership through community decision-making, not books.

Achievement: Organized villagers to seize stored grain and distribute it among the hungry. When the British tried to arrest him, thousands stood by his side.

Impact: Remembered as the "Lion of Chhattisgarh" — the man who proved food could be a weapon of justice against oppression.



Inspiring Stories : Thillaiaadi Valliammai

Thillaiaadi Valliammai

Background: Born in 1898 in Johannesburg to Tamil immigrant parents.

Struggle: Faced racial discrimination in South Africa and injustice under oppressive pass laws.

Education: Learned activism by joining Gandhi's non-violent campaigns.

Achievement: Went to prison at just 16 for protesting against racist laws, refusing to accept bail or back down.

Impact: Her sacrifice inspired South African Indians to join the struggle in larger numbers, showing that courage knows no age.



Inspiring Stories : Ajijan Bai

Ajijan Bai Background: Born in Lucknow in the 1800s, known as a skilled mujra dancer.

Struggle: Saw her land taken, people oppressed, and culture insulted by the British.

Education: Learned arts, etiquette, and sharp observation skills from court life.

Achievement: Secretly gathered British military secrets during performances in 1857. Led a 400-woman force called “Mastani Toli.” Refused riches, asked for a soldier’s uniform. Captured and executed by being tied to a cannon.

Impact: Became a symbol of how brains and bravery can destroy even the strongest enemy plans.



Inspiring Stories : Kanaklata Barua

Kanaklata Barua

Background: Born in 1924 in Assam's Gohpur area, from a humble farming family.

Struggle: Lost her parents at a young age, worked to support her siblings.

Education: Limited schooling, but learned patriotism from local youth groups.

Achievement: At 17, led a procession with the national flag towards a police station despite being told to turn back.

Impact: Became a youth icon in Assam, remembered for her determination to lead when most would have stepped aside.

Error handling

- Error detection
- Error correction

ED :- we know that an error has occurred but dont know where

EC :- we know that an error has occurred also where it occurred



ED → Error

EC → III bit


Buffer
Packet loss

EH

ED

1) D + D

RT

X Traffic X 2 ✓

! in a million

X 2) Parity : even odd

Powerful

3) CRC
4) Checksum
} CN

EC

1) Hamming code X CN

Digital logic

C R C → check.

Cyclic Redundancy

⊕ → ex-or.

		⊕
0	0	0
0	1	1
1	0	1
1	1	0

⊕ → modulo two sum

$1 \oplus 1 = 0$

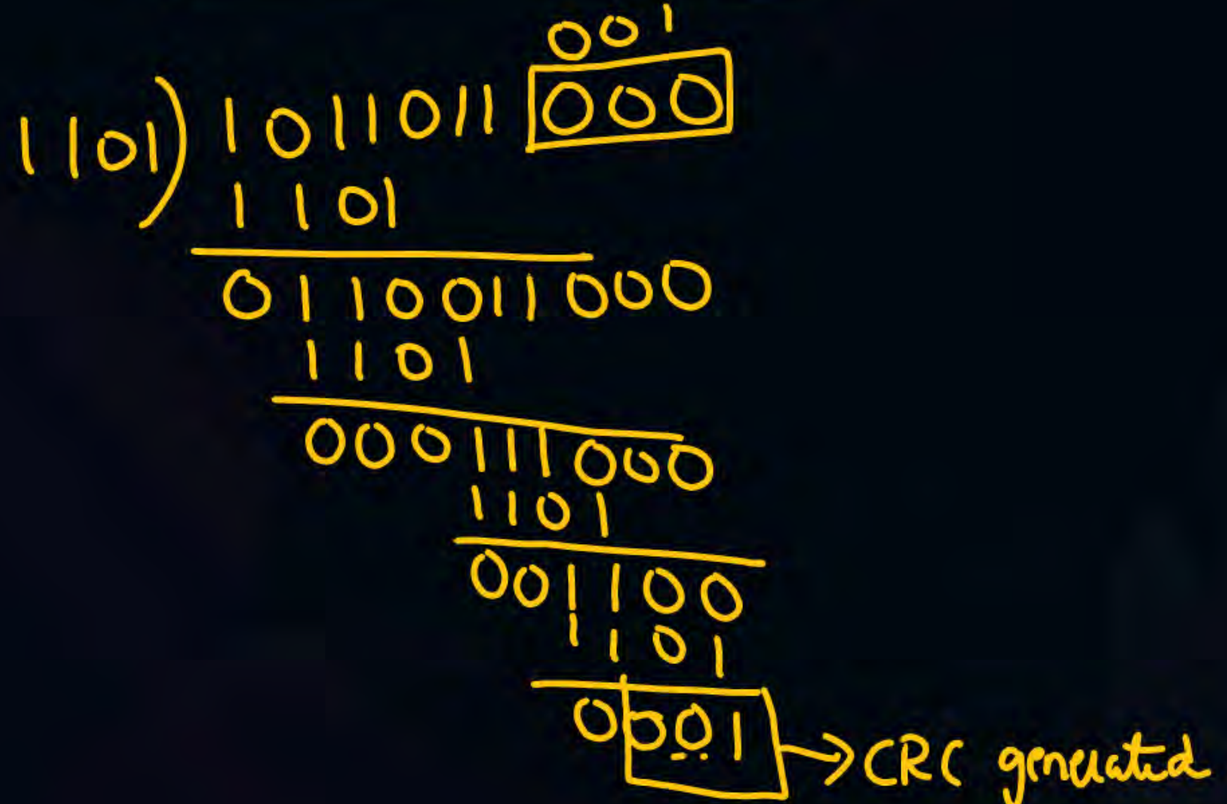
$1 \oplus 1 \oplus 1 = 1$

$1 \oplus 0 \oplus 1 = 0$

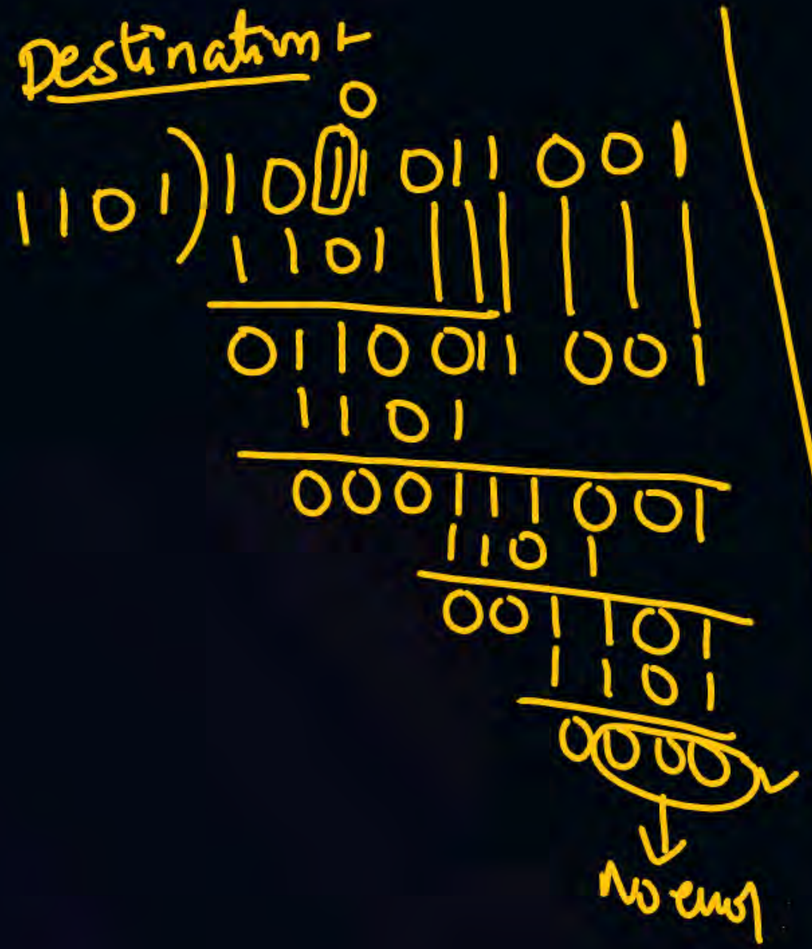
$1 \oplus 1 \oplus 1 \oplus 1 = 0$

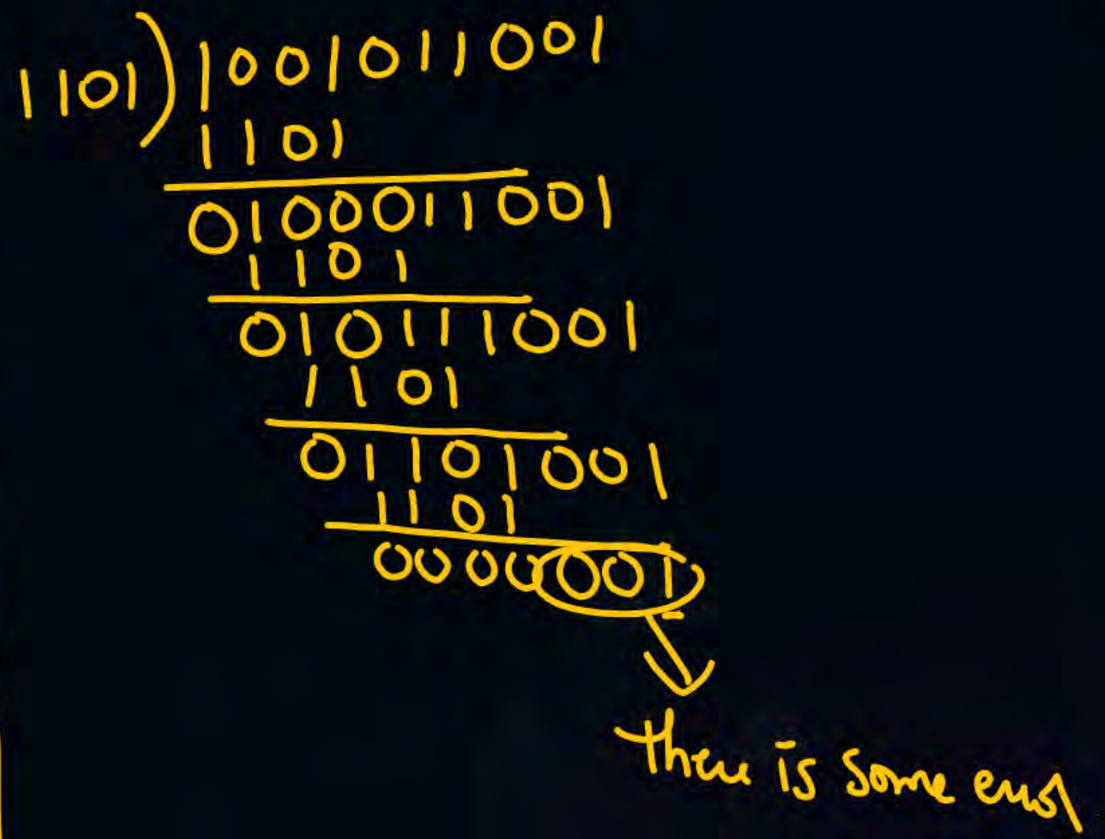
Source ✓

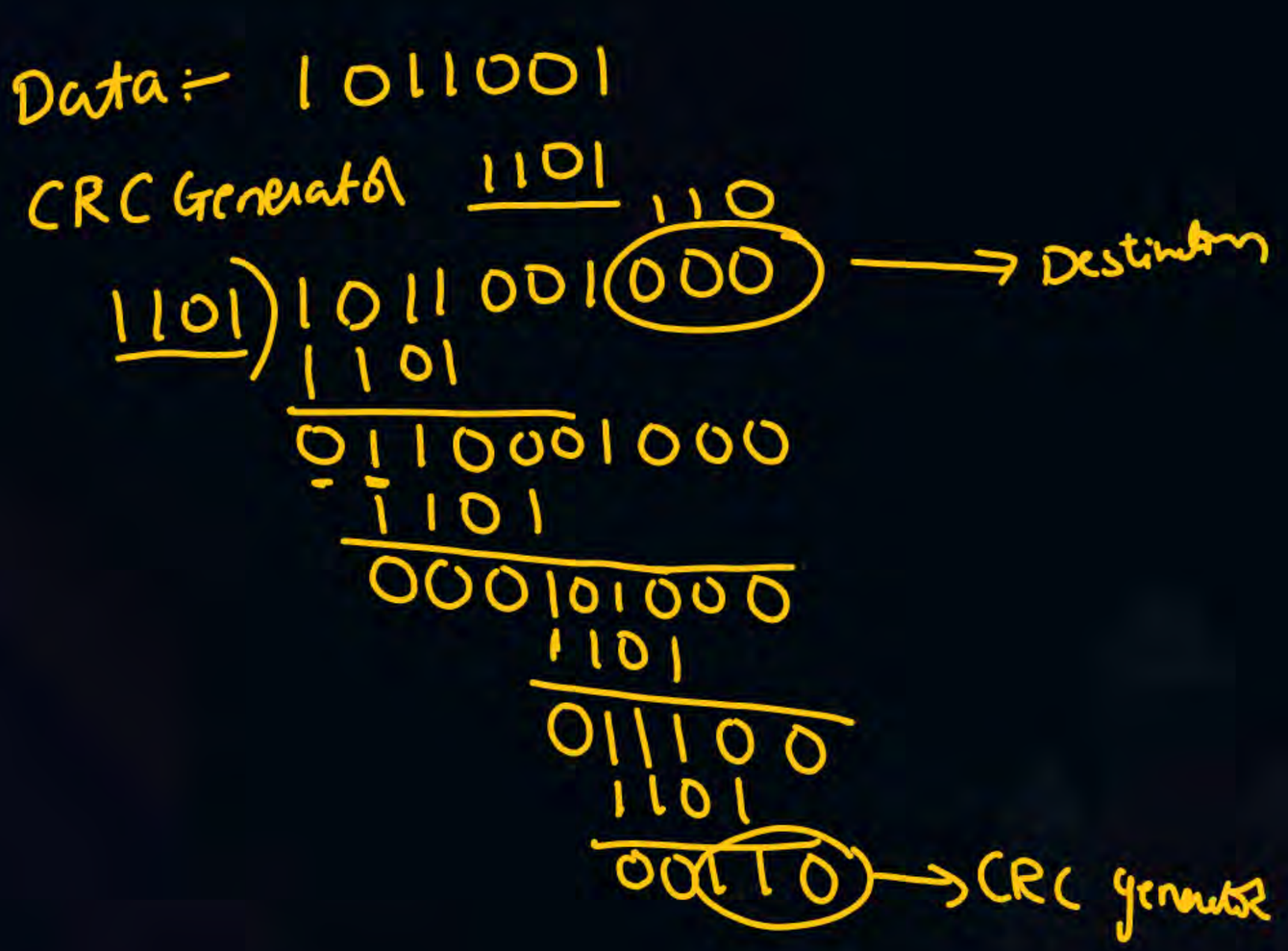
Data: 1011011 CRC Generated = 1101 → 3 zeros



Destination






Data :- 1011001
CRC Generator 1101
110
1101) 1011001000
1101
0110001000
1101
0001010000
1101
0111000
1101
00110
CRC generator
Destination


1101) 10011 001 110
1101
0110 0011 10
1101
0001 0111 0
1101
0110 10
1101
000
No em
1101) 1001 001 110
1101
0100 001 110
1101
0101 011 10
1101
0111 1110
1101
0010 110
1101
0110 0
1101
001
→ end.

Sometimes CRC generator will be given as a polynomial

$x^3 + x + 1 \Rightarrow 1 \cdot x^3 + 0 \cdot x^2 + 1 \cdot x + 1 \cdot x^0 = 1011$ ✓

CRC Gen = n

(n-1) 0's

Data - 11010(000)

```
      1011
    ------------
0110 0000
     1011
    ------------
0111 1000
     1011
    ------------
0101 1000
     1011
    ------------
0010 1000
     1011
    ------------
0001 0000
     1011
    ------------
0000 0010
```

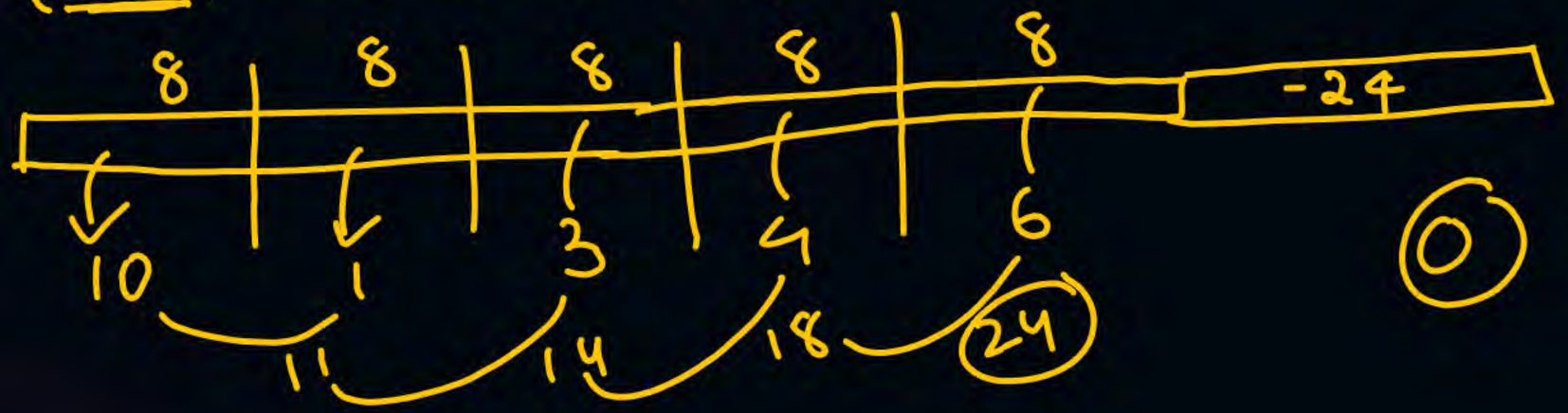
CRC generated ← 010

Data - 11010 1000 ✓

```
      1011
    ------------
0110 1001 000
     1011
    ------------
0111 1100 000
     1011
    ------------
0100 0000 000
     1011
    ------------
0010 0000 000
     1011
    ------------
0001 0000 000
     1011
    ------------
0000 0000 111
```

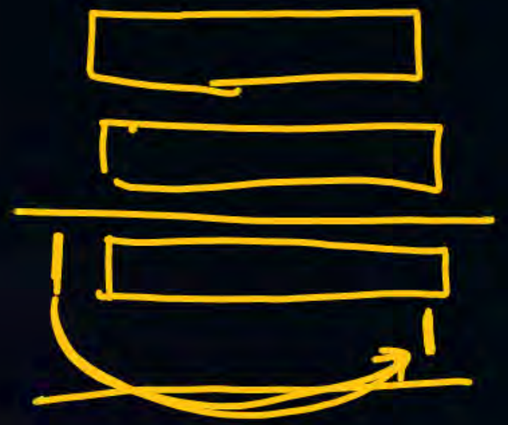
→ CRC generated

Checksum :- (8 CS, 16 CS, 32 CS, 64 CS)

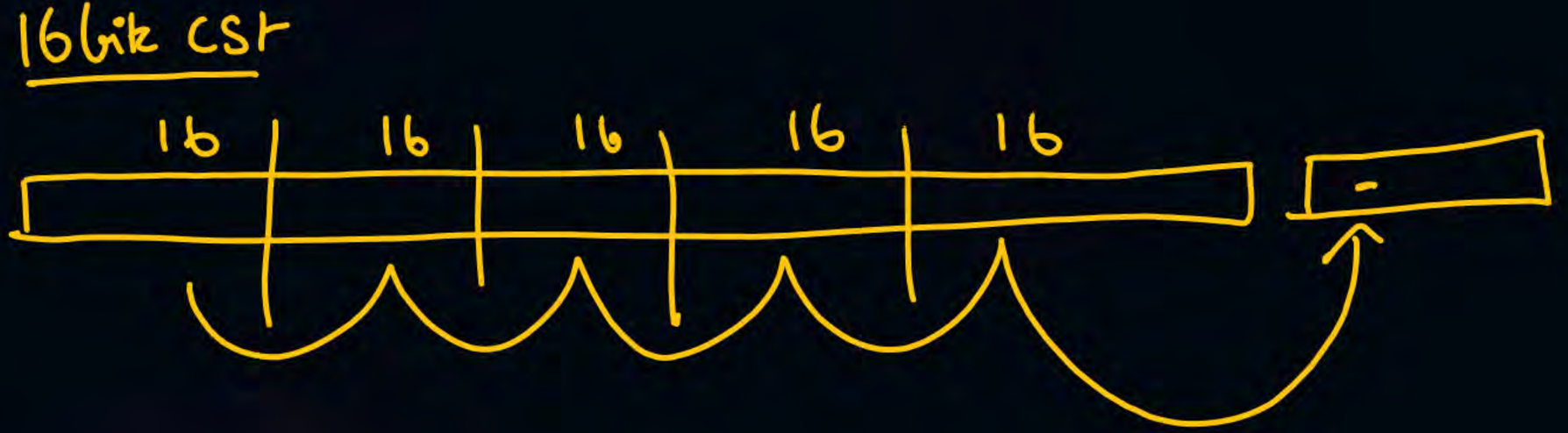


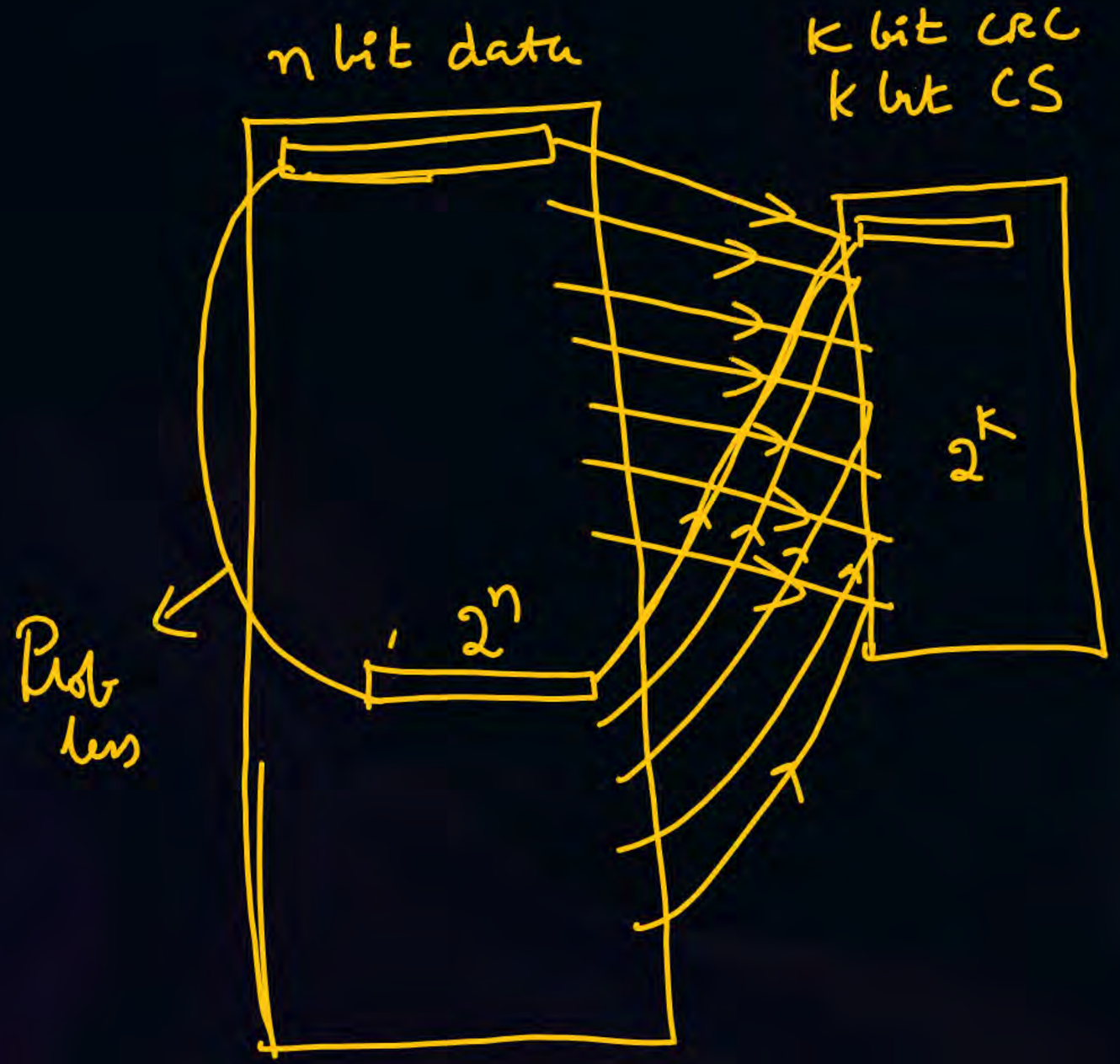
(8) bit checksum





End around carry

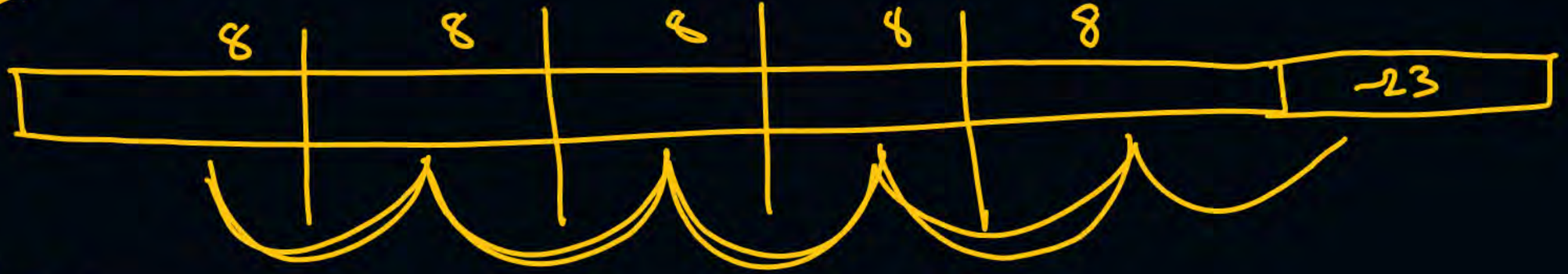

16bit CSA
16
16
16
16
16



DLL – CRC

NL, TL – CS

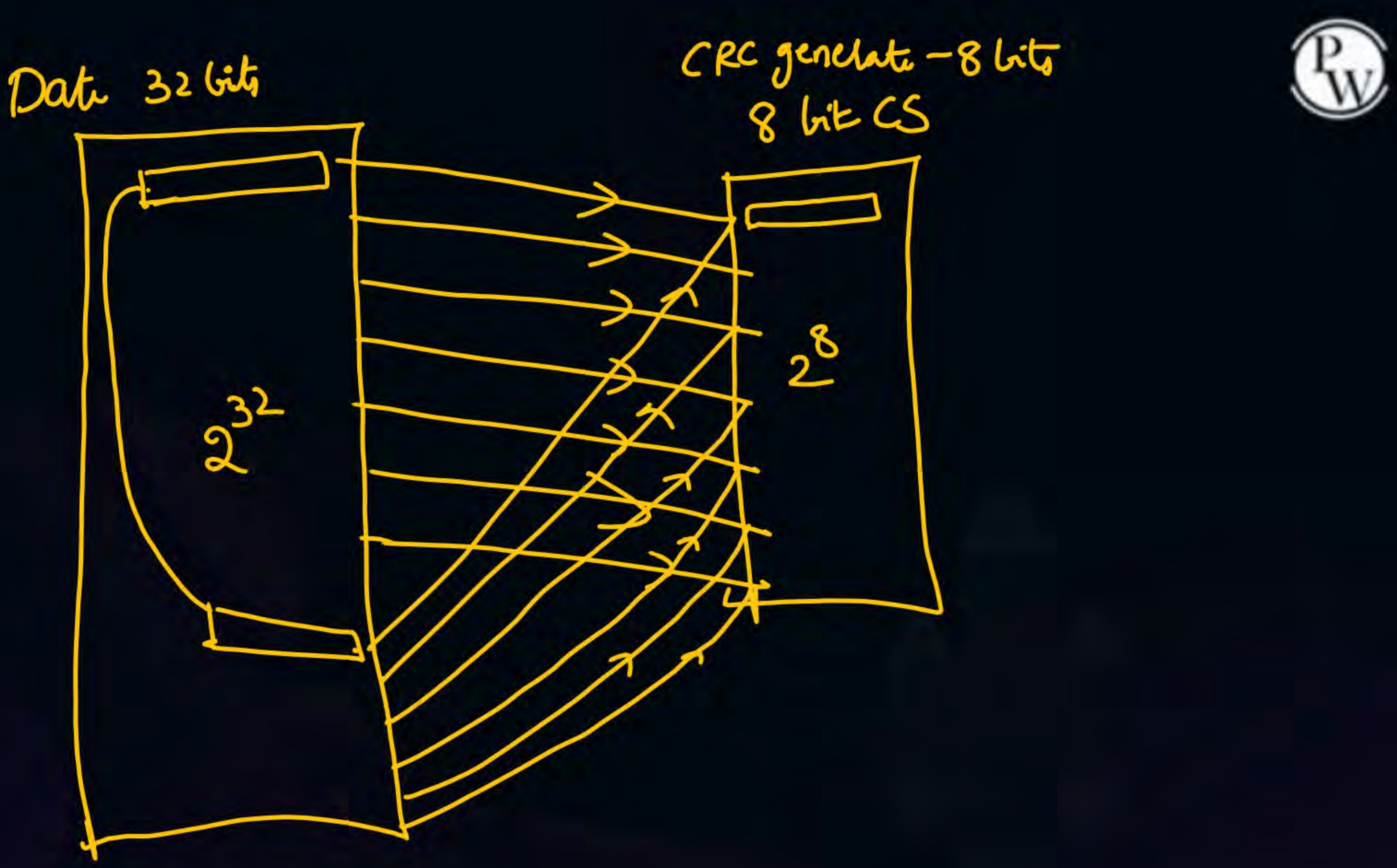
Sender 8 bit CS

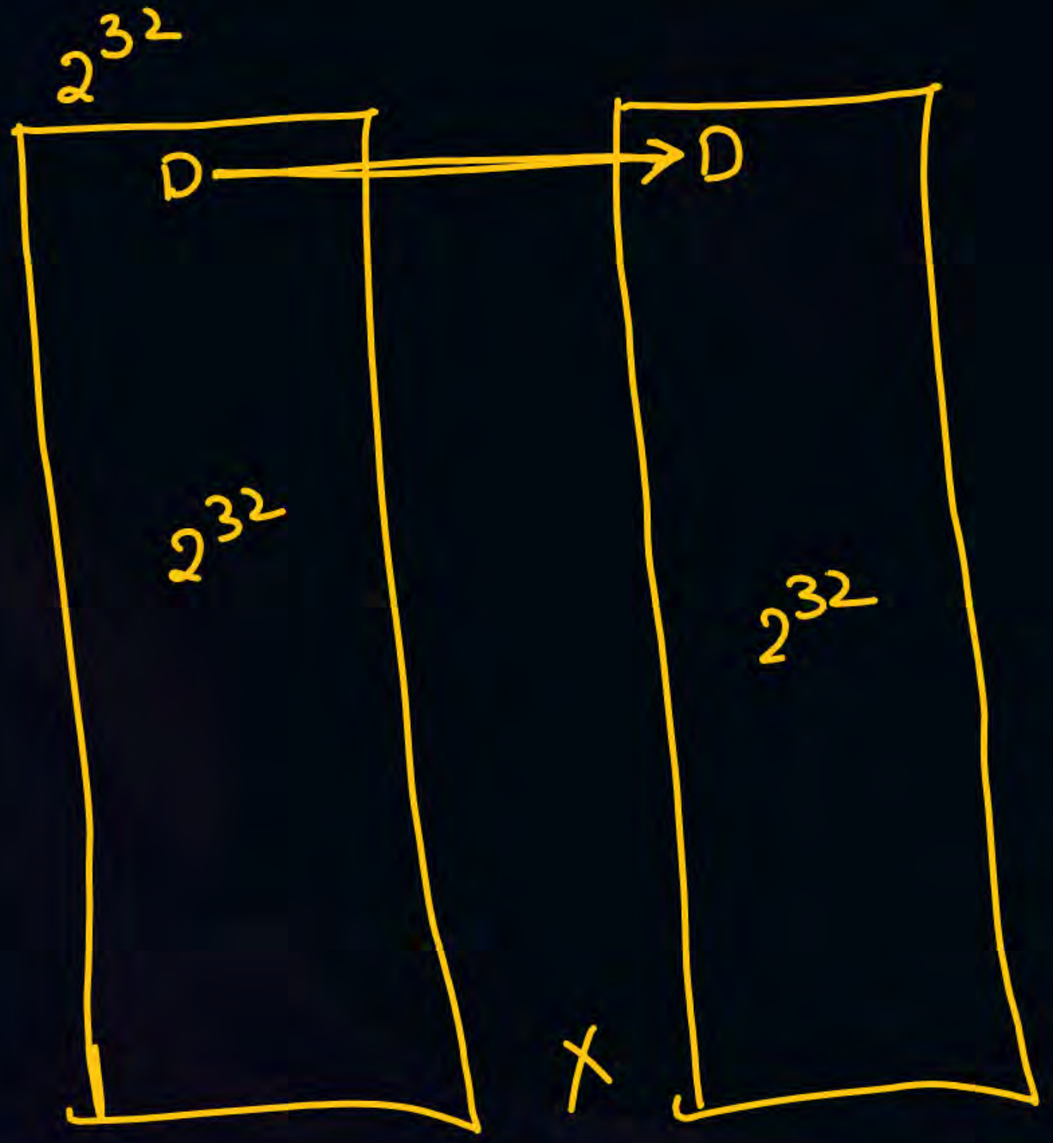


Recv

23

16 bit CS, 32 bit CS '0'


Data 32 bits
CRC generate - 8 bits
8 bit CS
2^{32}
2^{8}


2^{32}
D
D
2^{32}
2^{32}
X

Purpose of CN



IPC



CN H/W and S/W

functions of CNS:-

functions

mandatory

1) EC
2) FC
3) AC
4) mul & demul



ready

70 funding

optional

1) Encry / decryp ✓
2) checkpointing ✓
3) Routing ✓

.....

These 70 functions are grouped in diff reference models

Like
1) ISO-OSI
2) TCP/IP
3) ATM
4) X.25
5) IEEE
.
.
.
.
.
.

ISO — OSI

ISO
- International
- Standard
- Organization

OSI
- Open
- Systems
- Interconnect

ISO - OSI :-

- Application layer
- Presentation layer } user oriented
- Session layer
- Transport layer — Thick
- Network layer — Complex
- Data link layer — H/w & SW
- Physical layer — H/w

Physical layer

It deals with electric, mechanical, functions and procedural characteristics of physical link

- If the physical link is copper wire, then message is converted to electric signal
- If the physical link is optical fibre, then message is converted to light
- If the PL is wireless, then message is converted to electromagnetic waves

Transmission modes:-



Simplex — Radio, TV antenna



half duplex — walkie talkie



full duplex — (Cell phone), (Computer)

Topologies :



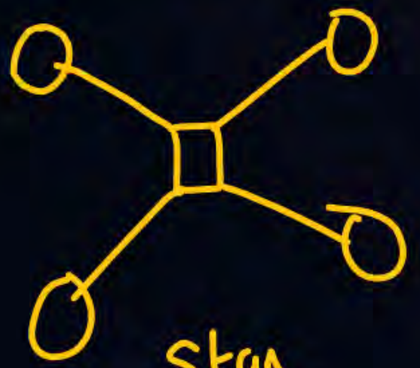
Bus



Ring



mesh



Star



Tree

Encoding :-

manchester and differential manchester encoding

manchester :-



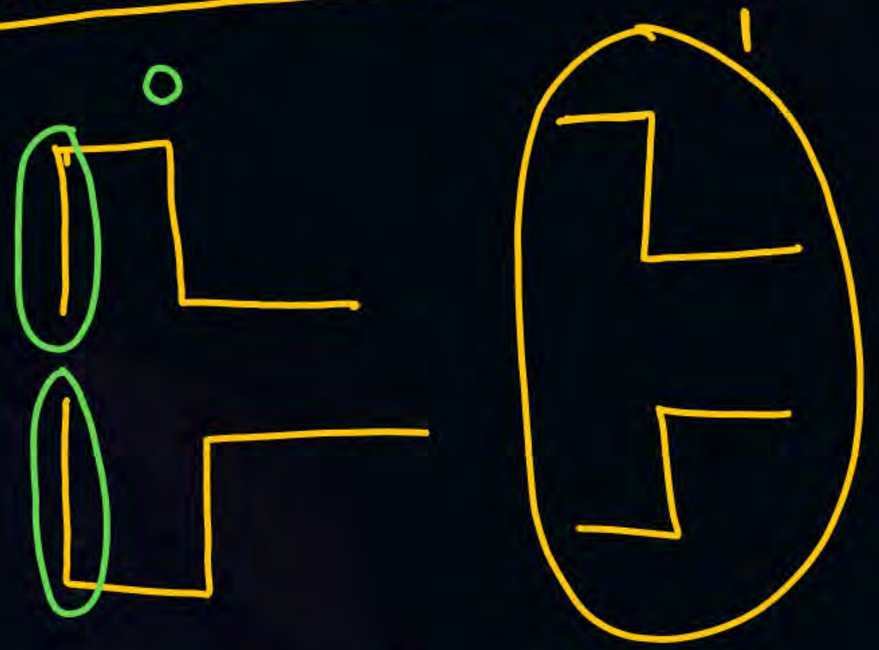
manchester encoding:-



2007 ✓



Differential manchester encoding :-









2007

Gate:-

Baud rate = 2 * bit rate

↓

voltages changing per sec

m and dm

DLL :-

1) Flow control (S&W, GBN, SR)

   DLL uses GBN

2) Error control (CRC, CS)

   DLL uses CRC

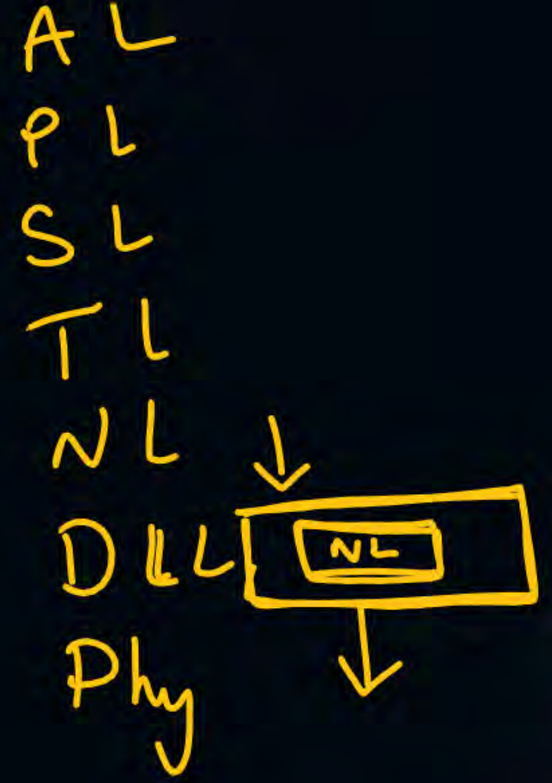
3) Framing

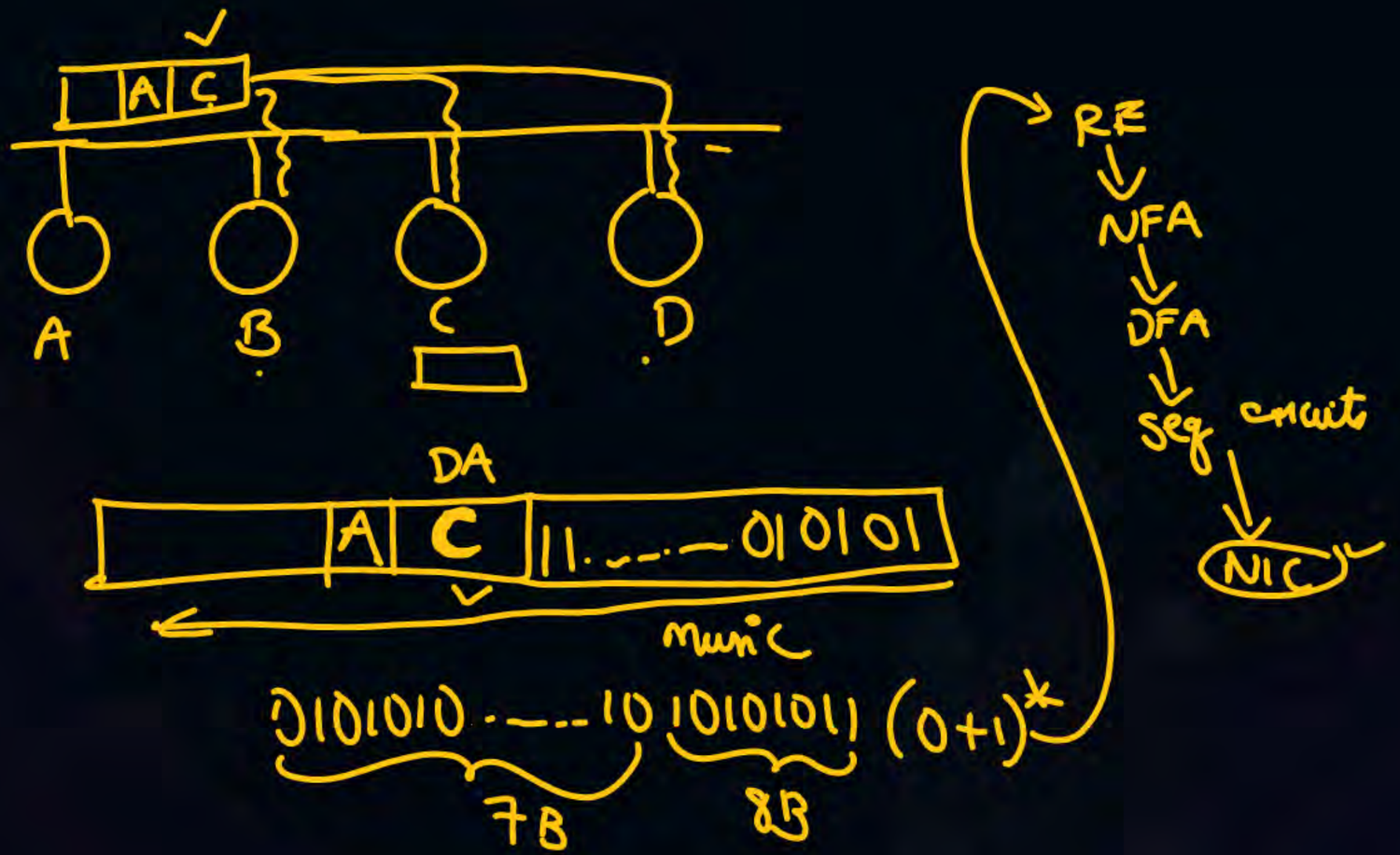
4) Physical addressing

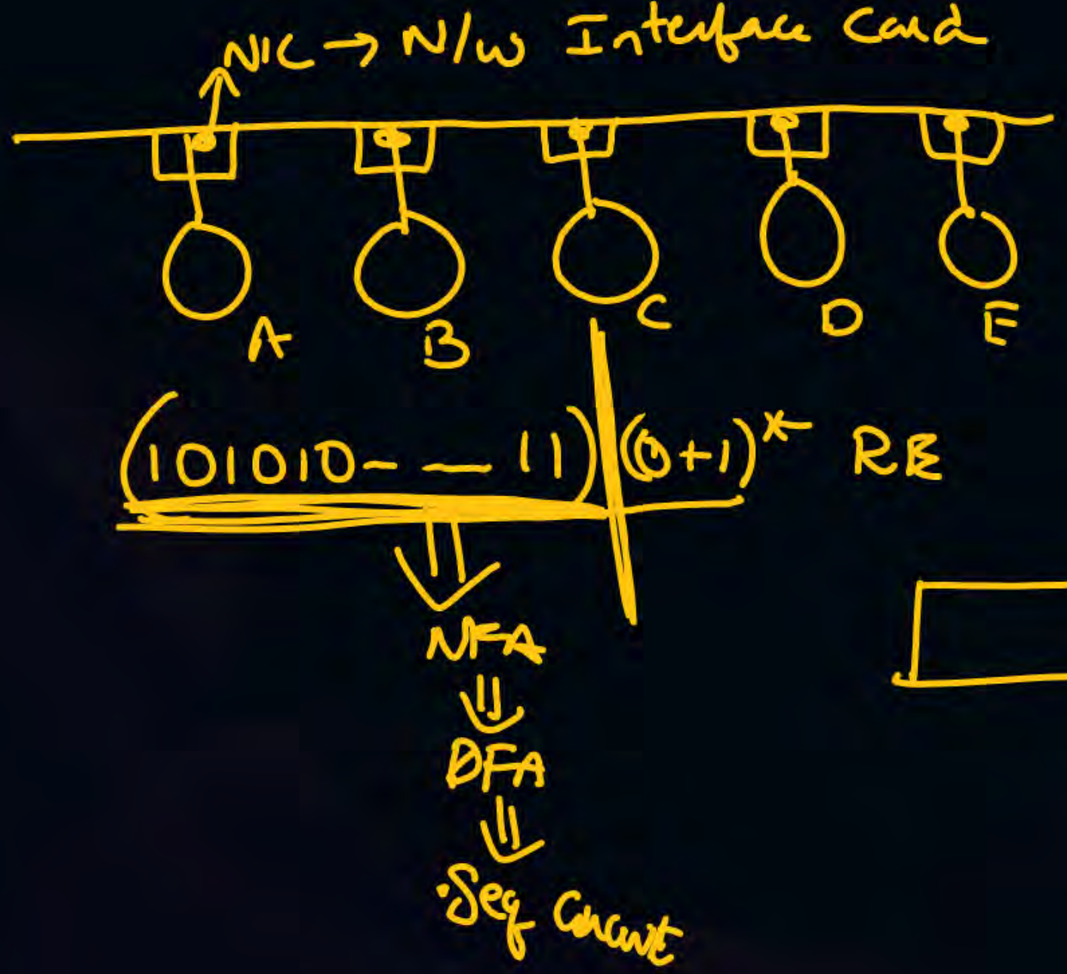
5) Access control (CSMA/CD)

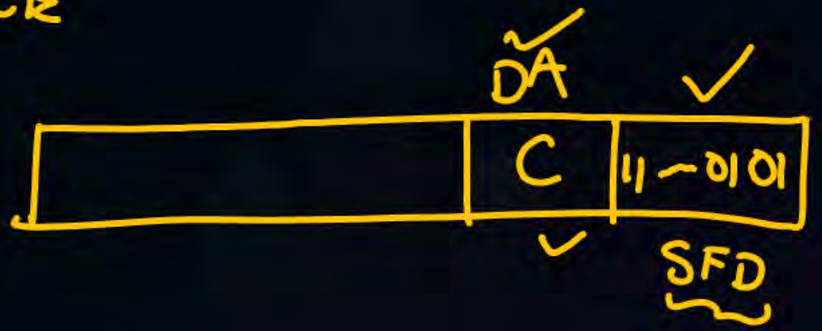
Framing :-

Framing :- Taking data from NL and putting it in a frame and giving it to PL for transmission




A C
A
B
C
D
DA
A C
11.....01 01 01
music
0101010.....10 10101011 (0+1)*
7B
8B
RE
NFA
DFA
Seq
circuit
NIC


NIC → N/w Interface card
A
B
C
D
E
(101010- — 11)(0+1)* RE
NFA
DFA
Seq Circuit


DA
C
11-0101
SFD

In general, in LANs we use multi access channel.
All the hosts are connected to a single channel.



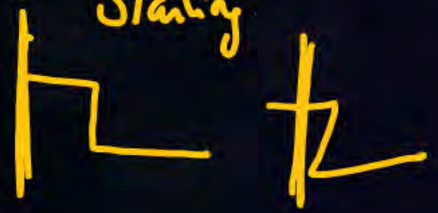
Any station can receive message at any time, so all the stations have to know the beginning of a message. For this, we use SFD

Start Frame delimiter



SFD → indicate that packet is starting

Also SFD is used for synchronization.



THANK - YOU